

Project Report: AIE Project

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Introduction

This project is a Streamlit-based web application designed for data processing, visualization, statistical analysis, and generating a linear trend neural network. The goal of the system is to offer a clean, intuitive interface for exploratory data analysis (EDA), feature transformation, and prediction model output with export functionality.

1 Project Architecture

The project is structured into separate modules for clarity.

Main files:

- **app.py** – main Streamlit interface.
- **data_loader.py** – data loading and initial preprocessing.
- **plots.py** – plotting functions.
- **transforms.py** – log transform and outlier capping.
- **ml.py** – neural network forecast model.
- **stats_tests.py** – implementation of the t-test.
- **export.py** – exporting results to JSON.
- **constants.py** – global variables.

2 Functionality

The Streamlit app provides these key analytical tools:

2.1 Data Upload

User can upload their own dataset or rely on a default file path, which already contains AirQuality.csv file. The module performs basic validation and initial cleaning.

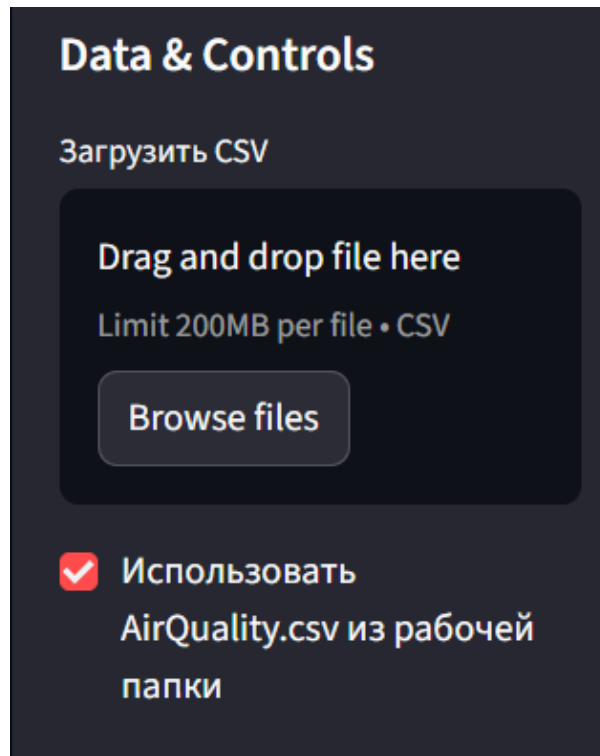


Figure 1: Upload Interface

2.2 Visualization

The app includes:

- Boxplots of distributions,
- Trend plots for time series,
- Correlation heatmaps.

2.3 Data Transformations

The following transformations are supported:

- Logarithmic feature transformation,

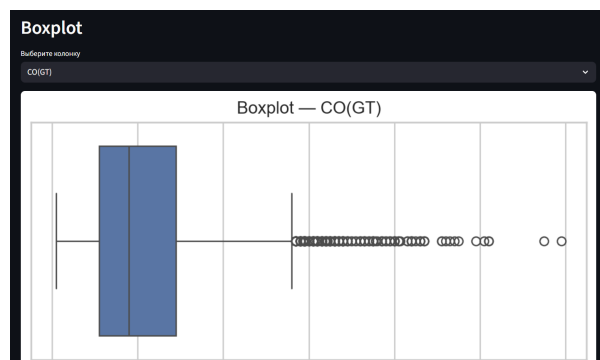


Figure 2: Boxplot

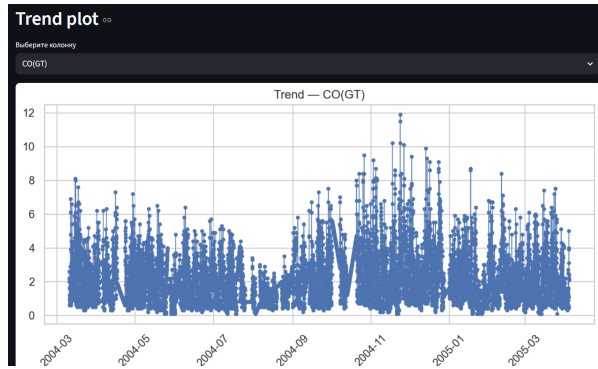


Figure 3: Trend

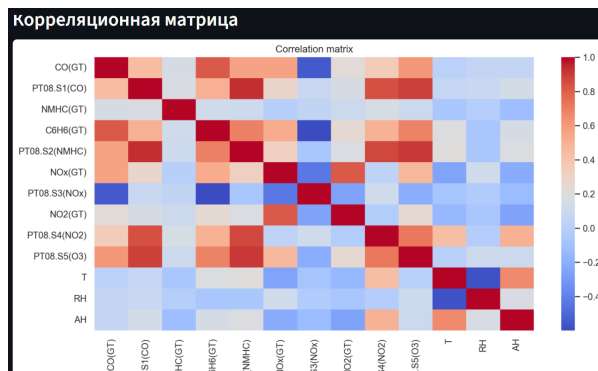


Figure 4: Heatmap

- Outlier capping.

2.4 Statistical Analysis

A classical two-sample t-test is implemented.

2.5 Forecast Model

A linear regression simple neural network is used to generate trend-based predictions.

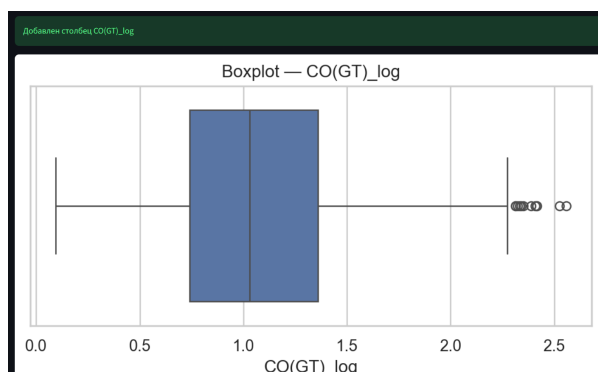


Figure 5: Log Transform

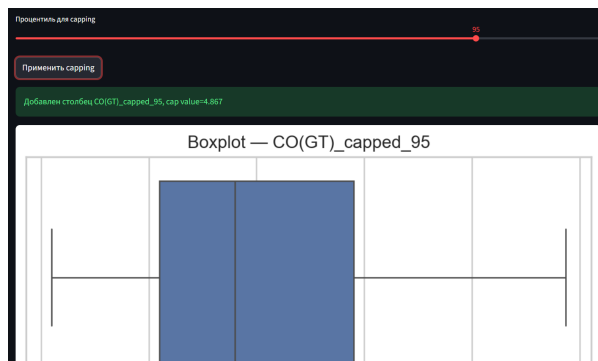


Figure 6: Outliers

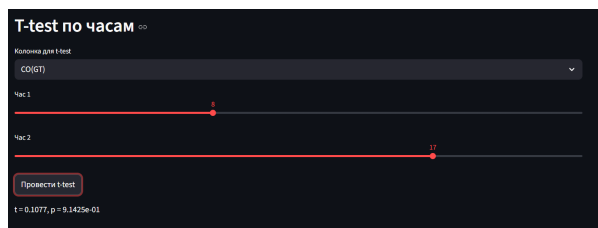


Figure 7: T-test

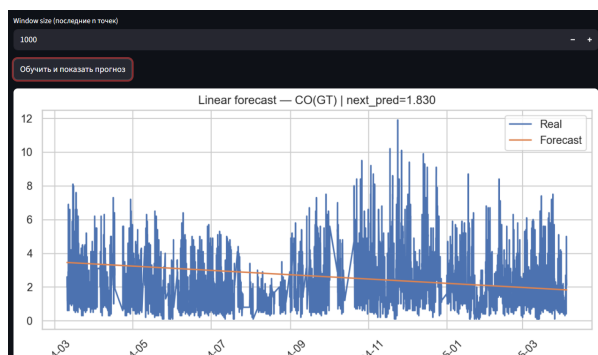


Figure 8: ML

Air Quality interactive analysis														
Описательная статистика (после очистки) ср														
	Datetime	CO(GT)	PT08.S1(CO)	PM10(GT)	COHS(GT)	PT08.S2(NMHC)	NOx(GT)	PT08.S3(NOx)	NO2(GT)	PT08.S4(NO2)	PT08.S5(O3)	T	RH	
count	9357	9357	9357	9357	9357	9357	9357	9357	9357	9357	9357	9357	9357	93
mean	2004-09-21 16:00:00	2.1306	1048.9963	-159.0901	10.1792	894.5953	168.617	794.9902	58.1489	1391.4796	975.072	18.2334	49.15	
min	2004-03-10 18:00:00	0.1	-200	-200	0.1	-200	-200	-200	-200	-200	-200	-1.9	-1	
25%	2004-06-16 00:00:00	1.1	921	-200	4.5	711	50	637	53	1185	700	11.7	3	
50%	2004-09-21 16:00:00	1.8	1053	-200	8.3	895	141	794	96	1446	942	17.6	4	
75%	2004-12-28 03:00:00	2.9	1221	-200	14.1	1105	284	960	133	1662	1255	24.3	6	
max	2005-04-04 14:00:00	11.9	2040	1189	63.7	2214	1479	2683	340	2775	2523	44.6	8	
std	None	1.4317	329.8327	139.7891	7.5038	342.3333	257.4339	321.9936	126.9405	467.2101	456.9382	8.7818	17.15	

Figure 9: EDA stats

2.6 EDA Analysis

3 Methods and Formulas

3.1 T-test

The implemented formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \quad (1)$$

3.2 Linear Forecast

The linear regression model minimizes:

$$\min_{a,b} \sum_{i=1}^n (y_i - (ax_i + b))^2 \quad (2)$$

The predicted future value is given by:

$$\hat{y} = ax_{future} + b \quad (3)$$

4 Conclusion

The project provides a full pipeline for exploratory data analysis, visualization, statistics, and neural network forecasting. Its modular structure makes the code base easy to maintain and extend.